

Experimental study on the performance enhancement of PV/T by adding graphene oxide in paraffin phase change material emulsions

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ABSTRACT

Traditional water-based PV/T systems have low heat storage density, high PV backsheet temperatures, and low energy utilization in practical applications. In this work, nanoparticles were applied to improve thermal conductivity and simultaneously emulsions were added to enhance energy storage density of the conventional heat transfer media. Furthermore, the synergistic effect of nanocomposite Phase Change Material Emulsions (PCME) as the heat transfer medium on the thermoelectric efficiency of PV/T was studied in a real system. Specifically, the water-based 30 % paraffin emulsions with graphene oxide (GO) concentration of 0.01, 0.02 and 0.03 wt% were prepared by ultrasonic stirring and their thermophysical properties were characterized. The results show that the 30 wt% paraffin emulsion with addition of 0.02 wt% GO has the phase change temperature of 35 °C, latent heat of 42.93 J/g, thermal conductivity of 0.505 W/(m·K) and viscosity coefficient of 0.91. It shows the thermal efficiency of the paraffin/GO emulsion increased by 92.28 %, the electrical efficiency increased by 8.87 %, the comprehensive efficiency increased by 45.75 %, and 15.8 % increase in heat collection for the heat exchange tank in comparison with the water. It is concluded application of PCME would improves the thermoelectric performance of the system than traditional water-based PV/T systems.

1. Introduction

The widespread use of renewable energy is an effective mean of combating the energy shortage, achieving carbon peaking and carbon neutrality, in which solar energy attracts considerable interest due to its abundant and renewable. The PV/T systems have received considerable attention in recent years due to their eco-friendly energy applications and usage in various domains such as hot water production [1], heating [2], desalination [3], and agricultural drying [4]. In practice, system architecture and heat transfer medium are the main factors affecting the thermoelectric efficiency of PV/T systems [5]. In traditional solar PV/T systems, water [6,7] and air [8] are commonly employed as heat transfer mediums. However, both air and water recover heat by sensible heat with low thermal conductivity. The thermal efficiency of a system utilizing water as the heat transfer medium ranges from 30 % to 70 %, while its electrical efficiency varies from 7.58 % to 13.8 %. And the

thermal efficiency of a PV/T system utilizing air as the heat transfer medium ranges from 23 % to 56 %, and its electrical efficiency varies from 7.7 % to 15 % [9].

Nanofluids and Phase Change Material Emulsions (PCMEs) are new heat transfer mediums for PV/T systems that are currently being studied. Nanofluid is the addition of nanoparticles with high thermal conductivity to a heat-carrying fluid (e.g., water), which can form micro-convection currents in the fluid and facilitate the heat transfer process, thus improving the thermal conductivity of the heat-carrying fluid [10]. The use of water-based nanofluids in PV/T systems can effectively reduce the temperature of PV panels and thus increase the output efficiency of PV modules [11]. Sheikholeslami et al. [12] reviewed the effect of different types of water-based nanofluids on the performance of flat plate solar collectors (FPSCs) and concluded that among the nanofluid-based FPSCs, 24 % of the nanoparticles were Al₂O₃, 12 % were multi-walled carbon nanotubes (MWCNT) and the rest were other

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types of nanoparticles. At the optimal nanoparticle additions, Al_2O_3 and MWCNT resulted in a maximum increase of 28.8 % and 30 % in the thermal efficiency and 8.8 % and 17 % in the electrical efficiency of the system performance, respectively. After examining the effect of different water-based nanofluids on solar PV/T systems by experimental studies and simulation analysis, Sangeetha [13] and Abdo [14] concluded that nanofluids containing carbon nanoparticles exhibit superior heat transfer performance compared to other types of nanofluids.

However, water-based nanofluids have been found to have low energy storage density and limited ability to recover latent heat. Therefore, PCMEs have been introduced as latent heat storage fluids. PCMEs are two-phase mixed fluids, consisting of a dispersed phase, i.e., PCMs, an interface layer (surfactant), and a base fluid, typically water. Specifically, PCMEs are prepared by dispersing PCMs into water in the presence of surfactants [15,16]. At present, some researchers mainly focus on the simulation of PCME heat transfer and thermal storage process to study its energy storage density, and there is a lack of analysis and research on the effect of practical application of PCME. Liang et al. [17] carried out a numerical simulation study on the heat transfer and storage characteristics of PCME in a layered heat storage tank. Their findings suggested that compared with water, PCME can increase the system heat transfer by 60 %, and the energy storage capacity is 1.6–1.7 times that of water. Similarly, Shao et al. [18] numerically investigated the flowability and heat transfer features of paraffin emulsion in the finned-tube heat exchanger. Their results indicated that the pumping consumption of the emulsion is lower than that of water at the same heat transfer rate and that the heat transfer rate of the emulsion is 1.1–1.3 times that of water at low flow rates (0.3–0.6 m/s). Morimoto et al. [19,20] simulated the heat transfer characteristics of PCME under different flow states in a circular tube and discovered that regardless of laminar or turbulent flow, the convective heat transfer coefficient of the emulsion would rise during the phase change process. Wang et al. [21] studied the thermal management performance of PCME in a lithium-ion battery pack by using water and PCME as coolants, respectively. In investigating the influence of different coolant flow rates on the thermal storage capacity through simulation, they found that the maximum temperature of the system with 10 wt% emulsion was 1.1 °C lower than that of the water-based system under the same conditions. The simulation results indicated that the cooling capacity of the coolant rose with the increase of the flow rate.

For improving the thermal conductivity and energy storage density of heat transfer media at the same time, nanoparticles and PCMEs were added. Zhao et al. [22] prepared paraffin/GO (0.1–0.3 wt%) emulsions by ultrasonic emulsification and examined the influence of GO on the properties of paraffin emulsion. It was concluded that the thermal conductivity of the paraffin emulsion rose and then fell with increasing GO mass fraction, reaching a maximum of 0.645 W/m·K at 0.2 wt%. a 109.42 % increase in thermal conductivity over pure paraffin emulsions. Xiang et al. [23] incorporated graphene nanoparticles in a 20 wt% paraffin emulsion to form a composite PCME with low supercooling and high thermal conductivity using high-speed shearing (8000 rpm). It was discovered that 1.2 wt% of graphene reduced the supercooling to 0.27 °C and increased the thermal conductivity by 20.22 %. Soliman [24] and Murray [25] investigated the impact of nanoparticles with varying concentrations on heat transfer characteristics during the melting of organic PCMs. They found that an increase in the mass fraction of nanoparticles exerted a positive effect on enhancing the heat transfer efficiency of PCMs and reducing the melting time. Miqdam et al. [26] employed nano TiO_2 as a thermal conducting particle for water and paraffin in their research, which revealed that 2 wt% TiO_2 nanoparticles increased the thermal conductivity of water and paraffin by 126.6 % and 170 %, respectively. Regarding the research on nanocomposite PCMEs, most of the studies have focused on the preparation work and the characterization of emulsion properties with the addition of nanoparticles. This has led to the development of nanocomposite PCMEs with high energy storage density and high thermal conductivity, while there

are still some research gaps in the application of nanocomposite PCMEs as heat transfer medium in practical PV/T systems.

The aim of this study is to further improve the thermoelectric efficiency of PV/T systems using nanocomposite PCMEs. Only a few works have investigated the practical application of PCMEs in solar systems and most of the PCMEs used are Micro-encapsulated Phase Change Materials (MPCMS) slurry [27,28] and pure paraffin emulsion [29]. The MPCM slurry preparation process is complicated, prone to leakage and there are some different phase changes between the outdoor experiments and indoor experiments. Therefore, based on pure paraffin emulsion, the 30 % paraffin emulsions with GO of 0.01, 0.02 and 0.03 wt % was prepared and then applied in a solar PV/T system to study its impact on the thermal efficiency, the electrical efficiency and comprehensive efficiency. With this experiment, on the one hand, it can verify the performance enhancement effect of nanocomposite PCME on the actual solar energy system, thus facilitating its promotion in the field of solar energy storage, and on the other hand, it can provide data support for the simulation work of nanocomposite PCME in the system of heat transfer and storage process.

2. Experimental section

2.1. Preparation and characterization of paraffin/GO emulsions

Water-based Paraffin emulsion with Diluted concentration of 44 wt% possessing a phase change temperature of 35 °C was purchased from Shanghai Yijiu Chemical Co., LTD. GO with microsheet size of 0.2 ~ 10 μm and a peelable rate ≥ 97 % were obtained from Shenzhen Suiheng Technology Co. LTD. Beyond that, the instruments involved in the preparation of paraffin/GO emulsion are universal balance, mixer and ultrasonic cell pulverizer. The model of universal balance: LS120A-SCS, with the range of 0–120 g; the model of mixer: JJ-1B precision constant speed (120 W) small electric mixer; the model of ultrasonic cell crusher: SCIENTZ-IIID (1000 W), with the crushing capacity of 0.5–600 ml.

As is shown in Fig. 1, initially, graphene oxide powder is weighed and added to deionized water to form a uniformly dispersed aqueous suspension via blender stirring and ultrasonic cell pulverizer ultrasonic dispersion for 25 min under the power of 450 W. Moreover, water-based paraffin emulsion with the concentration of 44 wt% is diluted into 30 %. Subsequently, the homogeneously dispersed GO aqueous suspensions is slowly added into the water-based emulsion with 30 wt% paraffin while being constantly stirred at a speed of 1200 rpm for 40 min. The mixture is then ultrasonically dispersed for an additional 25 min to obtain paraffin/GO emulsions with graphene oxide concentrations of 0.01, 0.02 and 0.03 wt% and paraffin concentrations of 30 wt%.

The experimental setup used for dynamic stability testing consists of a water bath and a refrigerator. The water bath is used to heat the paraffin/GO emulsions, while the refrigerator is used to cool them. The cycling conditions for the stability testing involve a temperature interval of 25 °C to 60 °C and back to 25 °C, with the number of cycles set at 10,

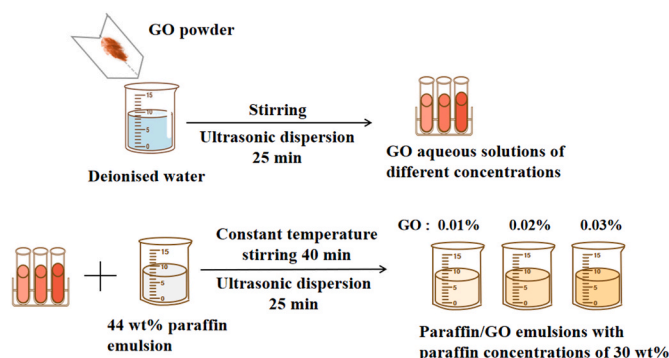


Fig. 1. Paraffin/GO emulsions preparation process.

30, and 50. The thermal properties of the emulsions, including the melting and crystallization peaks, onset temperature of phase change, and heat change during the phase change, are determined using the DSC-200PC thermal flow differential scanning calorimeter provided by NETZSCH (temperature accuracy: $<0.1^\circ\text{C}$, enthalpy accuracy: $<0.1\%$). The temperature range of DSC test is $-20\text{--}70^\circ\text{C}$, the temperature rise and fall rate is 10 K/min , and the temperature is constant for 2 min after reaching the specified temperature each time. Thermal conductivity tester type TC-3000 (measurement range & accuracy: $0.0005\text{--}5\text{ W/(m}\cdot\text{K)}$, $\pm 2\%$) for testing the thermal conductivity of emulsions, which is capable of measuring the thermal conductivity of various liquids and solid substances, with a minimum sample dosage of 30 ml for liquid samples. Viscosity measurements are carried out using the NDJ-8S digital display viscometer (measurement range & accuracy: $1\text{--}2000000\text{ mPa}\cdot\text{s}$, $\pm 3\%$). The viscosity test uses the viscous moment generated by the rotor in the emulsion to react to the viscosity of the emulsion. For the test, 0# and 1# rotors are selected for each group of samples for multiple tests, in which 1# rotor is used for the paraffin emulsion with the addition of 0.03 % of GO, and the rest of them are used for 0# rotor. In order to improve the accuracy of the data, in the thermal conductivity and viscosity tests, the average value is taken from every 5 sets of test data.

2.2. Set-up of solar PV/T systems

To study the influence of nanocomposite PCME on the thermoelectric performance of the PV/T system, water is used as a control group. In this regard, two flat-plate heat pipe solar PV/T systems are constructed and labeled as System 1 and System 2. The schematic diagram of a flat-plate heat pipe solar PV/T system is shown in Fig. 2, which is composed of two major sections: the collector circulation system and the electric energy output system. The collector circulation system is made up of a solar collector (as depicted in Fig. 3), a heat exchange water tank, a make-up water tank, and a circulation pump. In this system, the heat generated by the PV backsheets is collected and transported to the heat exchange tanks through heat pipes and ducts (medium is emulsion or water). In particular, the PV panels are monocrystalline solar photovoltaic panels supplied by Xuzhou Capita Energy Co., Ltd, with dimensions of $1580\text{--}810\text{--}40\text{ mm}$ and an effective heat collection area of 1.28 m^2 . The heat pipes are flat heat ones, filled with a small amount of acetone as the working medium, with 12 heat pipes evenly arranged per photovoltaic panel. The pipes are made of corrugated stainless steel pipes DN 25 and wrapped in polyurethane insulation foam of 25 mm thickness. The heat exchange water tanks have a capacity of 160 L. The electrical output system consists of a solar controller, a battery, and a data acquisition device. The angle and connection of the components for both systems

are identical. The experimental setup is presented in Fig. 3, and both systems are erected on the roof of the Experimental 2 Building of Beijing University of Architecture, facing due south at an angle of 37° with the horizontal ground.

A experimental system is equipped with a solar radiometer, various temperature sensors, and a solar controller. Hourly solar radiation intensity is measured with the solar radiometer, while temperature sensors collect data on the PV backsheet temperature, heat pipe temperature, and water temperatures at both the inlet and outlet of the joint tank, as well as the heat exchange tank temperatures. Additionally, the solar controller measures the voltage and current output from the PV panels.

2.3. Data processing

The thermoelectric performance of PV/T systems is assessed by two indicators of thermal and electrical efficiency [30]. The electrical efficiency η_{pv} is the electrical power output of solar collector divided by the total amount of solar energy absorbed by the solar collector, which can be expressed by equation (1).

$$\eta_{pv} = \frac{q_{pv}}{\zeta G A_c} \quad (1)$$

where, q_{pv} is the instantaneous electrical power of the solar collector, W; G is the intensity of solar radiation, W/m^2 . A_c is the area of the PV panel surface, m^2 ; ζ is the PV cell coverage factor, i.e. the area of the PV panel cells as a proportion of the collector's heat gathering area, here $\zeta = 0.9$.

The q_{pv} is the product of the PV module output voltage U (V) and current I (A), which can be expressed by equation (2), where both voltage and current values are derived from actual measurements.

$$q_{pv} = UI \quad (2)$$

The thermal efficiency η_t is the thermal yield (thermal power) of solar collector divided by the total amount of solar energy absorbed by solar collector, which can be expressed by equation (3).

$$\eta_t = \frac{q_t}{G A_c} \quad (3)$$

where, q_t is the instantaneous thermal power of solar collector, W.

The calculation of the thermal power of System 1 needs to be divided into three types of states, namely in before phase change, during phase change and after phase change. The instantaneous thermal power q'_t before and after phase change of the system 1 medium can be expressed as the equation (4)

$$q'_t = c_1 \rho_1 v (t_{out} - t_{in}) \quad (4)$$

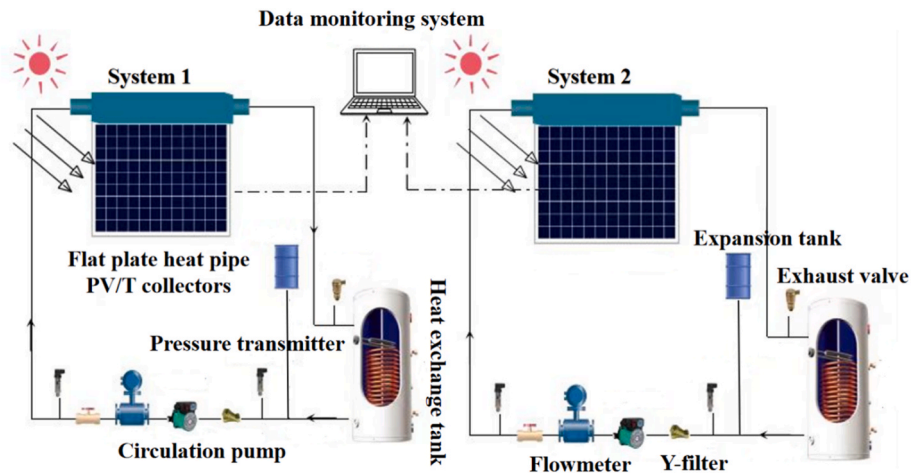


Fig. 2. The schematic diagram of solar PV/T system.

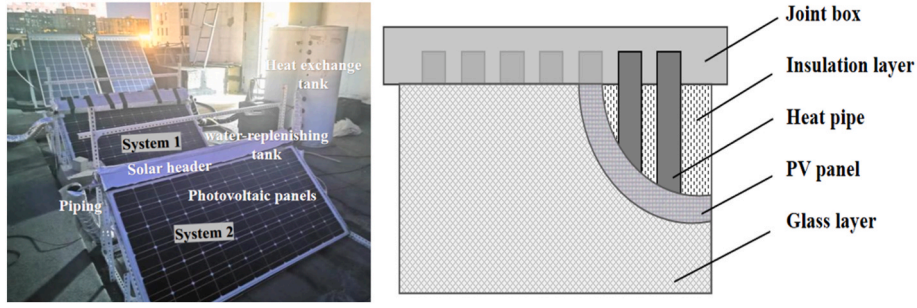


Fig. 3. Solar PV/T system test bench and solar collector structure diagram.

where, c_1' is the specific heat capacity of the emulsion before and after the phase change, J/(kg·°C); ρ_1' is the density before and after the phase change of emulsion, kg/m³; v is the volume flow rate of the heat transfer medium. The volume flow rate set for this experiment is 3 L/min, i.e. $v = 5 \times 10^{-5}$ m³/s; t_{in} and t_{out} are the inlet and outlet temperatures of the coupled box, °C, which are derived from experimental measurements.

The instantaneous thermal power q_t' during the phase change of the system 1 medium is given by Equation (5) as:

$$q_t' = v \rho_1' r \quad (5)$$

where, ρ_1' is the density of emulsion during phase change, kg/cm³; r is the latent heat of emulsion, kJ/kg. The latent heat of emulsion is taken from the DSC test data.

To fully characterize the paraffin/GO emulsion and water in different states, the specific heat capacity and density of the three states of the emulsion are calculated. The emulsion in the experiment is mainly composed of paraffin, water and a small amount of GO. The mass fraction of paraffin, water and GO can be approximately 30 %, 69.8 % and 0.02 %, and the specific heat capacity of the emulsion can be calculated based on the mass fractions and specific heat capacities of the pure substances. Because the proportion of GO is so small, it can be ignored in the calculation. Before phase change, paraffin exists as small solid particles in the emulsion, and the specific heat capacity of solid paraffin is 3200 J/kg·°C [31], therefore $c_1' = 3200 \times 0.3 + 4200 \times 0.7 = 3900$ J/kg·°C. The density of solid paraffin is 900 kg/m³ [32], then $\rho_1' = 900 \times 0.3 + 1000 \times 0.7 = 970$ kg/m³. After phase change, the paraffin in the emulsion is transformed from solid to liquid, and the specific heat capacity of liquid paraffin is 2900 J/kg·°C [31], therefore $c_1'' = 2900 \times 0.3 + 4200 \times 0.7 = 3810$ J/kg·°C. The density of liquid paraffin is 880 kg/m³ [33], then $\rho_1'' = 880 \times 0.3 + 1000 \times 0.7 = 964$ kg/m³. The density of paraffin emulsion is dynamically changing as they are in a solid-liquid state during the phase change, thus the average of the density of paraffin emulsion before and after the phase change is taken as the density value of paraffin emulsion during the phase change. Table 1 summarizes the physical parameters of the paraffin/GO emulsion and water in different

states. The instantaneous thermal power of system 2 is calculated in the same way as system 1 medium before and after phase change, except that the values of specific heat capacity and density are based on the physical parameters of water.

Ultimately, the performance of the PV/T system is characterized using the combined utilization efficiency of the system. As the electrical and thermal energy output of the system are two different forms of energy with variable energy grades, the electrical energy needs to be converted into thermal energy to calculate. The combined efficiency of the system η_{pvt} is given by Equation (6) as:

$$\eta_{pvt} = \eta_t + \zeta \frac{\eta_{pv}}{\eta_{power}} \quad (6)$$

where, η_{power} is the conversion factor of electrical energy to thermal energy in a typical thermal power plant, taken as 38 % [34].

The viscosity of paraffin/GO emulsions increases in relation to water, and this increase in viscosity leads to an increase in the pressure loss of the system, which means that more pumping power will be required during the operation of the system. The instantaneous power consumption P_F is given by Equation (9) as:

$$P_F = \Delta P \times S \times u \quad (7)$$

$$u = \frac{v}{S} \quad (8)$$

$$P_F = \Delta P \times v \quad (9)$$

where, P_F is the power consumption due to pipeline pressure losses, W; ΔP is the system pressure losses, Pa; S is the pipe cross-sectional area, m²; u is the media flow rate, m/s.

The energy consumption due to system pressure loss during the test cycle is calculated by Equation (10) as:

$$P = P_F \times t \quad (10)$$

where, P is the total system energy consumption, J; t is the system runtime, s.

2.4. Uncertainty analysis

The direct measurement error is mainly generated by the length, temperature, radiation intensity and flow measurement. The length is mainly measured by tape measure, and its measurement accuracy is 1-level accuracy. The thermocouple used for PV plate temperature and fluid inlet/outlet temperature measurement is calibrated by constant temperature water bath, and the measurement error is ± 0.2 K. The measurement error of the solar radiometer is ± 7 W/m². And the accuracy of the electromagnetic flowmeter is 1.5-level accuracy.

For those physical quantities that are indirectly measured, they can be found by using the error transfer equation. When the indirectly measured physical quantity Y is composed of n directly measured physical quantities ($X_1, X_2, \dots, X_n$), its functional relationship expression

Table 1

Physical parameters of the components in the emulsions.

Materials	Density/(kg·m ⁻³)	Specific heat capacity/(J·(kg·°C) ⁻¹)
Water	1000	4200
Solid paraffin	900	3200
Paraffin (Solid-liquid)	890	—
Liquid paraffin	880	2900
30%Paraffin+0.02%GO (before phase change)	970	3900
30%Paraffin+0.02%GO (during phase change)	967	—
30%Paraffin+0.02%GO (after phase change)	964	3810

is given by Equation (11) as:

$$Y = f(X_1, \dots, X_n) \quad (11)$$

The transfer equation for the uncertainty of the indirectly measured physical quantity Y is given by Equation (12) as:

$$U_Y = \sqrt{\left(\frac{\partial f}{\partial X_1} \Delta X_1\right)^2 + \left(\frac{\partial f}{\partial X_2} \Delta X_2\right)^2 + \dots + \left(\frac{\partial f}{\partial X_n} \Delta X_n\right)^2} \quad (12)$$

$$\frac{U_Y}{Y} = \sqrt{\left(\frac{\Delta X_1}{X_1}\right)^2 + \left(\frac{\Delta X_2}{X_2}\right)^2 + \dots + \left(\frac{\Delta X_n}{X_n}\right)^2} \quad (13)$$

where, U_Y is the uncertainty in Y; $\frac{U_Y}{Y}$ is the relative uncertainty.

According to Equation (3), the relative uncertainty of electrical efficiency can be expressed by Equation (15):

$$Y_{\eta_{pv}} = f(t_{pv}, G, A_c) \quad (14)$$

$$\frac{U_{Y_{\eta_{pv}}}}{Y_{\eta_{pv}}} = \sqrt{\left(\frac{\Delta t_{pv}}{t_{pv}}\right)^2 + \left(\frac{\Delta G}{G}\right)^2 + \left(\frac{\Delta A_c}{A_c}\right)^2} \quad (15)$$

According to Equation (3) and Equation (4), the relative uncertainty of thermal efficiency can be expressed by Equation (17):

$$Y_{\eta_t} = f(t_{out}, t_{in}, u, G, A_c) \quad (16)$$

$$\frac{U_{Y_{\eta_t}}}{Y_{\eta_t}} = \sqrt{\left(\frac{\Delta t_{out}}{t_{out}}\right)^2 + \left(\frac{\Delta t_{in}}{t_{in}}\right)^2 + \left(\frac{\Delta u}{u}\right)^2 + \left(\frac{\Delta G}{G}\right)^2 + \left(\frac{\Delta A_c}{A_c}\right)^2} \quad (17)$$

Within the parameter range of this experiment, the relative uncertainties of the electrical and thermal efficiencies of PV/T systems 1 and 2 are 3.3 %, 1.54 %, 8.48 %, and 8.48 %, respectively, which are all within the 10 % error range, indicating that the experimental data have good accuracy.

3. Results and discussion

3.1. Characteristics and thermal physical properties of paraffin/GO emulsions

3.1.1. Stability

The morphology of the prepared paraffin/GO emulsions is observed by placing them in test tubes, and their state is examined after 2 and 30 days of standing. The results are presented in Fig. 4(a) and Fig. 4(b), which indicate that the emulsions remain well-dispersed and consistent with their initial state after 2 and 30 days of standing at room

temperature. No phase separation or delamination is observed in either case.

Stability is a key factor to consider in phase change emulsions, and it can be divided into static stability and dynamic stability, the latter of which relates to the ability of emulsions to withstand multiple thermal cycles. The thermal properties of the emulsions after the thermal cycling are tested because the emulsions in the system are subjected to heat-absorbing and exothermic cycling for a long period of time during the actual application process. The main thermal properties of phase change emulsions that affect the thermoelectric efficiency of the system are the latent heat of phase change and the thermal conductivity. Therefore, the latent heat value and thermal conductivity are used as the two indexes to measure the dynamic stability of emulsion. Fig. 5 illustrates the changes in latent heat and thermal conductivity of the emulsions after undergoing multiple thermal cycles. The results indicate that there are only minimal changes in the latent heat and thermal conductivity of the four emulsions, suggesting that the emulsions maintain their original heat storage and thermal conductivity even after several thermal cycles, reflecting a good dispersion stability.

3.1.2. Phase change latent heat

Fig. 6 displays the DSC pattern for the endothermic and exothermic storage of the PCMEs. The heat flow curves of the four emulsions demonstrate a consistent trend with temperature, exhibiting two peaks during the temperature change corresponding to two phase changes in paraffin wax, namely solid-solid and solid-liquid phase change. Before the solid-liquid phase change, paraffin wax is suspended in water in the form of small solid particles, after reaching the phase change temperature, the small solid particles begin to absorb heat and liquefy until the phase change is complete. The peak temperatures during melting process for the emulsions show no significant difference, whereas during crystallization, the peak solid-liquid temperatures of the paraffin/GO emulsions with 0.02 wt% and 0.03 wt% GO are slightly delayed, indicating that the presence of GO slightly increased the supercooling of the sample. The latent heat of the emulsions can be determined by calculating the area under the endothermic/exothermic peak of the DSC curve. Since the experiment is conducted in October and the solar PV/T system only undergoes a solid-liquid phase change, the latent heat is only calculated for the second phase change process. The calculations for the latent heat of the emulsions are presented in Table 1. The results in Table 2 and Fig. 6 demonstrate that the incorporation of high thermal conductivity nanoparticles has negligible effect on the latent heat of the paraffin emulsions.

3.1.3. Thermal conductivity

The thermal conductivity measurements of the four emulsions at various temperatures are presented in Fig. 7. As depicted, the thermal

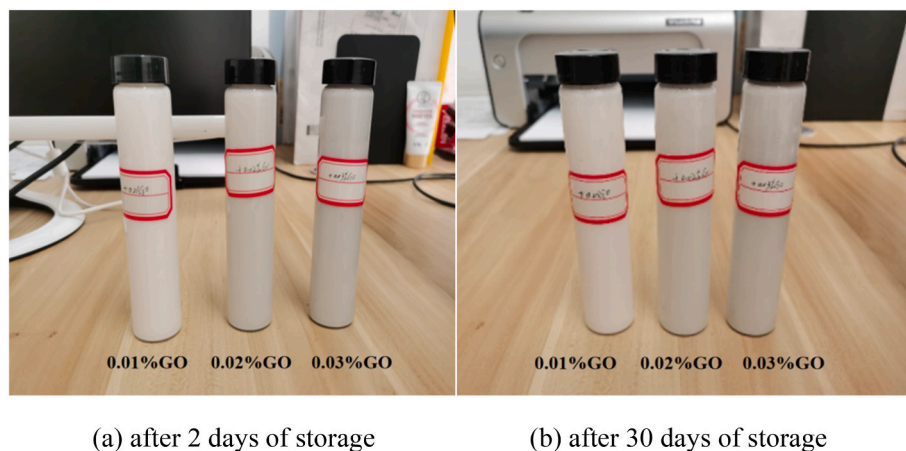


Fig. 4. Phase change material emulsions stability after standing.

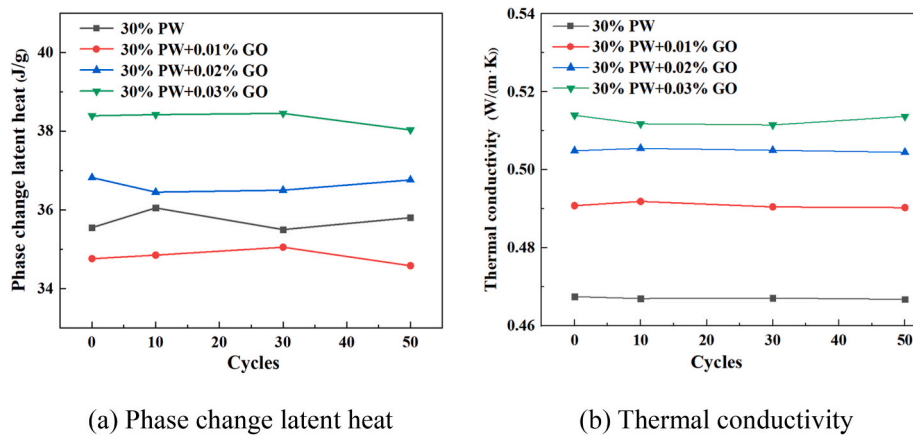


Fig. 5. Phase change material emulsions stability after thermal cycles.

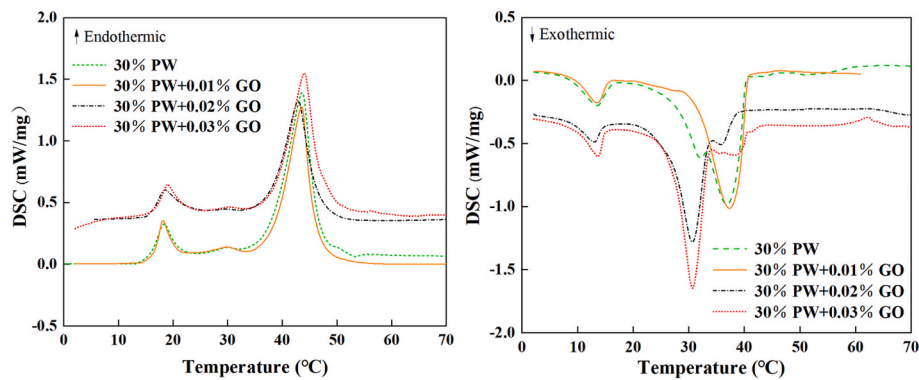


Fig. 6. (a) DSC spectra of phase change material emulsions in the endothermic phase; (b) DSC spectra of phase change material emulsions in the exothermic phase.

Table 2

Test data for phase change material emulsions.

materials	Heating process			Cooling process		
	First peak temperature/ °C	Second peak temperature/ °C	latent heat/ J·g ⁻¹	First peak temperature/ °C	Second peak temperature/ °C	latent heat/ J·g ⁻¹
30 % paraffin	18.5	43.1	35.55	13.6	37.2	41.25
30 % paraffin +0.01 % GO	18.2	43.6	34.76	13.6	37.3	40.05
30 % paraffin +0.02 % GO	18.5	42.9	36.82	13.0	30.7	42.93
30 % paraffin +0.03 % GO	19.0	44.0	38.39	13.6	30.6	39.88

conductivity of the emulsions exhibits a sharp rise once the temperature reaches 35 °C. This phenomenon can be attributed to non-uniform temperature distribution in the emulsions at the onset of phase change and the natural convection effect, which enhances the phase change rate of the emulsions. At 40 °C, the thermal conductivity drops sharply as the majority of the paraffin wax in the solution completes its phase transition, and the thermal conductivity of liquid paraffin wax is lower than that of solid paraffin wax. Furthermore, the temperature field distribution of the emulsions tends to be homogeneous, leading to a reduction in natural convection intensity. When the temperature reaches 45 °C, the phase change process of four emulsions ends, and all paraffin in the solution converts into liquid form. The enhancement in thermal conductivity can significantly enhance the thermoelectric performance of the solar PV/T system.

In addition, the emulsions thermal conductivity grew with rising mass fraction of GO nanoparticles in the paraffin emulsion. At 40 °C, the addition of 0.01, 0.02 and 0.03 wt% of GO nanoparticles result in a 5 %,

8 % and 10 % rise.

3.1.4. Viscosity

To enhance the effectiveness of the system in practice, it is crucial to maintain the fluidity of the PCMEs at a certain level as it directly impacts the resistance losses of the system. Lower resistance losses result in reduced pump consumption, which improves the efficiency of the system. Therefore, it is important to keep the viscosity of the emulsions at a minimum while ensuring that other properties are not compromised. The viscosity of the four emulsions was measured, and the results are shown in Fig. 8. The graph shows that the viscosity tends to drop as the temperature rises to around 70 °C, and after 70 °C, it would become stable at a certain value.

At the same shear rate, the viscosity η of the emulsions as a function of temperature T is as follows.

$$\eta = A \cdot \exp(B / T) \quad (18)$$

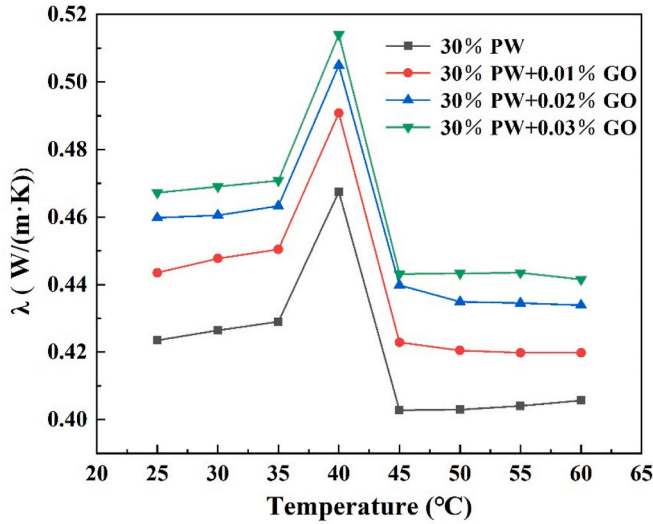


Fig. 7. Thermal conductivity of phase change material emulsions.

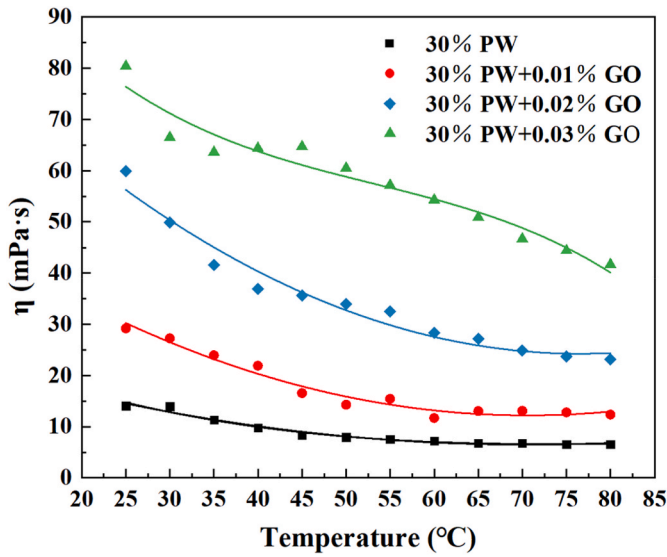


Fig. 8. The viscosity of phase change material emulsions.

where, A, B= Viscosity temperature coefficient, which is a certain constant. In this work, the magnitude of the A and B values are related to the type of PCMEs and the graphene oxide mass fraction, which can be calculated by fitting the Origin software, as shown in Table 3.

Table 3 indicates that the emulsions with 0.01 and 0.02 wt% GO are well suited for the system, while those with 0.03 wt% GO are slightly less suitable. The higher viscosity of the emulsion with 0.03 wt% GO can lead to adhesion of some paraffin wax to the rotor during the viscosity test, which can introduce significant errors. To avoid this, it is recommended to clean the rotor regularly during testing.

The comparison of the four emulsions in respect to thermal properties and fluidity suggests that the paraffin emulsion containing 0.02 wt%

Table 3
The value of viscosity-temperature coefficient.

Type of emulsions	A/(Pa·s)·K ⁻¹	B/(Pa·s)·K ⁻¹	R ²
30 % paraffin	4.29	31.75	0.94
30 % paraffin +0.01 % GO	8.35	33.22	0.92
30 % paraffin +0.02 % GO	19.14	24.71	0.91
30 % paraffin +0.03 % GO	36.93	20.48	0.86

GO exhibits superior overall thermal performance. Consequently, this specific paraffin/GO emulsion will be utilized for testing in a solar PV/T system.

3.2. Performance of PV/T system with paraffin/GO emulsion

To ensure the accuracy of the performance comparison experiments and to provide informative data, System 1 and System 2 were kept identical except for the heat transfer medium, while other equipment and components remained the same. However, in actual operation, there may be some inherent differences between the two systems. To clarify these differences, water was used as the heat transfer medium in both systems before conducting the performance comparison experiments using paraffin/GO emulsion and water as heat transfer mediums. Relative deviation values were calculated for each performance parameter after the operation of both systems.

The relative deviation is determined as given in Eq. (6).

$$RE = \frac{X_1 - X_2}{X_1} \times 100\% \quad (19)$$

where, X_1 = test results of system 1; X_2 = test results of system 2.

After performing the necessary calculations, the deviation values of the PV backplane temperature for the two systems are found to be within the range of -5.46 % to +5.49 %, with the deviation values being primarily concentrated around 2 %. Similarly, the deviation values of the heat pipe temperature for the two systems fall within the range of -5.42 % to +5.85 %, with the deviation values mainly being around 1 %. The deviation values of the heat transfer medium temperature are found to be within the range of -0.05 % to +2.37 %, whereas the deviation values of the water temperature in the heat exchange tank range from -1.82 % to +2.59 %. The electric efficiency deviation values vary from -5% to -7%, whereas the thermal efficiency deviation value falls within the range of +2.86 % to +8.54 %. Overall, the deviation values of the two systems are calculated to be within 10 %, hinting at the consistency between the two experimental systems used in this study. Therefore, it can be concluded that the two systems are almost the same, except for the difference in their heat transfer medium.

The experiment to compare the performance of the paraffin/GO emulsion and water system was conducted on October 11, 2021, from 09:00 to 17:00, during which data were collected at 3 min intervals between 09:00 and 17:00. The variations in the outdoor ambient temperature and solar radiation intensity on the test day are presented in Fig. 9. The average solar radiation intensity is determined to be 763.4 W/m², whereas the average outdoor ambient temperature is recorded as 22.5 °C.

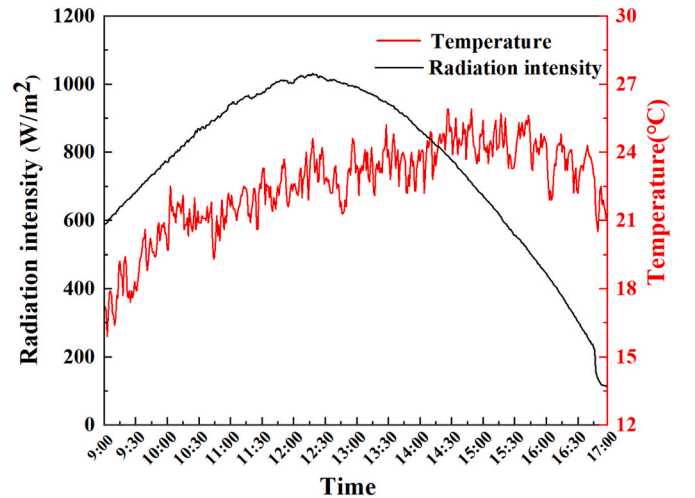


Fig. 9. Outdoor air temperature and radiation intensity.

3.2.1. Temperature

Fig. 10 illustrates the temperature variation of PV panel backsheet, heat transfer medium, heat pipe, and water in the heat exchange tank for both systems during the test period. The trend of PV backsheet temperature change is similar for both System 1 and System 2, rising and then falling, similar to the pattern of solar radiation intensity change. Notably, due to the high energy storage density of the paraffin/GO emulsion, the average PV backsheet temperature throughout the day in System 1 is approximately 2 °C lower than in System 2.

Given that the temperature of the heat transfer medium is constantly changing in the process of flowing through the joint box, the average of the inlet and outlet temperatures of joint box will be taken as the heat transfer medium temperature. Examining the change in medium temperature of the two systems in Fig. 10, it is found that the temperature of the heat transfer medium grows from approximately 25 °C–44 °C during the time from 9:00 to 15:00 in both systems. However, the medium temperature of System 2 (T_w) is consistently higher than that of System 1 (T_s), indicating that PCMEs absorb more heat with less change in temperature. The low temperature of the heat transfer medium reduces heat dissipation during system operation and prolongs the service life of the piping and equipment.

Regarding the heat pipe temperatures of the two systems, as is shown in Fig. 10, the difference between the heat pipe temperatures of the two systems is negligible before 11:30. After 11:30, the heat pipe temperature of System 1 is significantly lower than that of System 2. This is attributed to the fact that, before 11:30, the heat transfer medium of both systems flowing through the combined box and the condensing end of the heat pipe undergoes sensible heat transfer processes, absorbing almost the same amount of heat, and thus, the temperature difference between the two heat pipes is insignificant. However, when the emulsion in System 1 begins to undergo a phase change after 11:30, the heat transfer process between the emulsion and the condensing end of the heat pipe is latent heat transfer, during which the emulsion absorbs a large amount of heat. Consequently, the temperature of the heat pipe in System 1 is lower than that in System 2, as System 2 maintains sensible heat transfer all the time.

In terms of the trend of water temperature in the heat exchange tank, as indicated in Fig. 10, the temperature in both heat exchange tanks gradually rises as the systems run. The temperature in System 1 heat exchange tank is consistently higher than that in System 2 due to the energy storage properties of the paraffin/GO emulsion. After the experimental tests are completed, the temperature in the System 1 heat

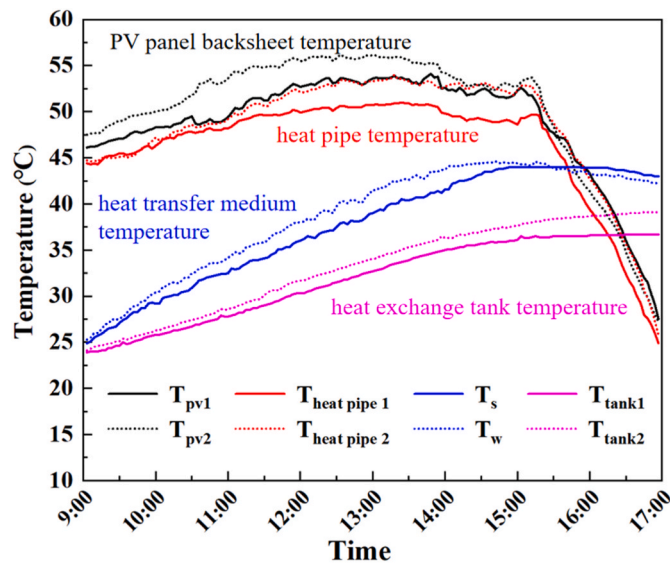


Fig. 10. The temperature change chart of photovoltaic backplane, heat pipes, thermal transfer mediums and heat exchange tanks.

exchange tank climbs from 24.3 °C to 39.1 °C, whereas the temperature in the System 2 heat exchange tank increases from 23.9 °C to 36.7 °C. However, as the volume of water in the tank is larger (160 L), it can be concluded that there is no significant difference in temperature between the two.

3.2.2. Thermal performance

Fig. 11 depicts the variation of thermal power and thermal efficiency of both systems during operation. It is evident from the graph that, prior to 11:30, the thermal power of system 1 is marginally lower than that of system 2. This can be attributed to the fact that, at this stage, both systems undergo heat transfer in the form of sensible heat, and the specific heat capacity and density of the paraffin/GO emulsion in system 1 are smaller than those of water in system 2. Following 11:30, the thermal power of system 1 substantially rises since the medium temperature of the system reaches 35 °C, and the emulsion initiates a phase change, leading to a substantial growth in its thermal conductivity. At 15:00 p.m., the emulsion essentially completes its phase change, leading to a momentary drop in the thermal power of system 1 due to its subsequent sensible heat exchange. However, once the phase change has been completed, the greater temperature difference between the heat transfer medium of system 1 and the condensing end of the heat pipe leads to greater heat exchange, resulting in system 1's thermal power remaining higher than that of system 2. During operation, system 1 can attain a thermal power of 1488 W, with an average thermal power of 427 W for the day, a 91.42 % increase over System 2. The thermal efficiency of system 1 follows the trend of the thermal power, with an average efficiency of 36.48 % for the day, a 92.28 % increase compared to system 2. Throughout the day, System 1 collects 9.95 MJ of heat, whereas System 2 collects 8.6 MJ of heat, resulting in a 15.8 % improvement in System 1 over System 2.

3.2.3. Electrical performance

Fig. 12 presents the changes in electrical power and electrical efficiency during the operation of the two systems. The graph illustrates that the electrical power of both systems grows from 9:00 to 12:30 and then declines after 12:30, following the trend of solar radiation intensity. System 1 has a higher average electrical power than System 2. This can be attributed to the paraffin/GO emulsion of System 1 absorbing more heat during heat exchange with the condensing end of the heat pipe, resulting in a lower temperature of the PV panels compared to System 2. The lower temperature of the photovoltaic panels contributes to lower internal resistance, higher photoelectric conversion efficiency, and higher electrical power output. System 1 achieves a maximum electrical power of 141 W and an average electrical power of

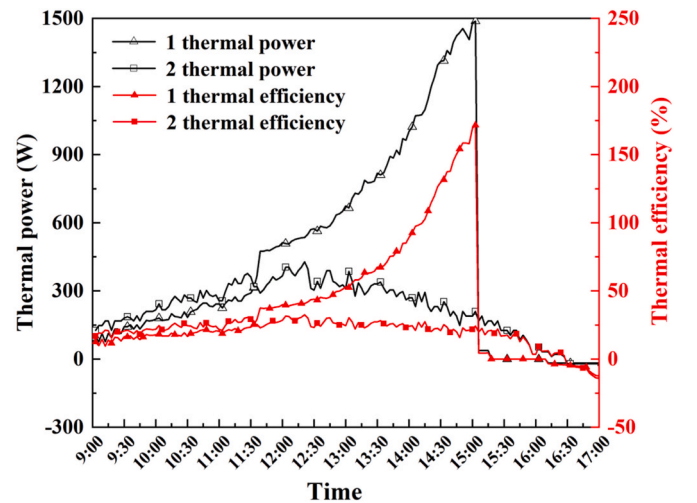


Fig. 11. The change chart of thermal power and efficiency.

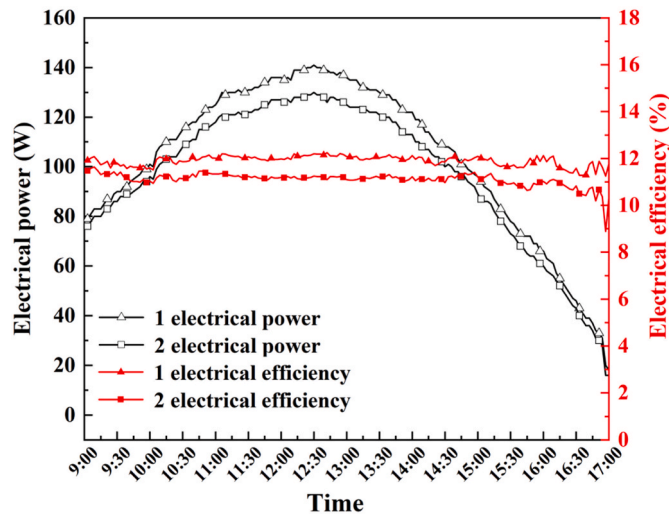


Fig. 12. The change chart of electric power and efficiency.

102.17 W for the day, while System 2 achieves a maximum electrical power of 130 W and an average electrical power of 95.39 W. System 1 exhibits a 7.2 % higher electrical power than System 2. The trend in electrical efficiency during the operation of both systems is similar, with a relatively stable efficiency from 9:00 to 16:30, followed by a significant drop after 16:30 as a result of the drop in solar radiation intensity. On average, System 1 exhibits a greater electrical efficiency than System 2, with an average electrical efficiency of 11.83 % for System 1 and 10.86 % for System 2, representing a 8.87 % increase in electrical efficiency for System 1 compared to System 2.

3.2.4. Comprehensive performance

Fig. 13 presents the changing trends of thermal efficiency, electrical efficiency and comprehensive efficiency during the operation of the two systems. Within this, the difference in thermal efficiency between the two systems is particularly noticeable, with System 1 having an average thermal efficiency of 38.36 % and System 2 an average thermal efficiency of 19.81 %, an increase of 92.28 % for System 1 over System 2; System 1 has an average electrical efficiency of 11.83 % and System 2 an average electrical efficiency of 10.86 %, an improvement of 8.87 % for System 1 over System 2. In addition, the change trend in the comprehensive efficiency of the systems is consistent with the changes in the thermal efficiency of the systems, due to the fact that the thermal

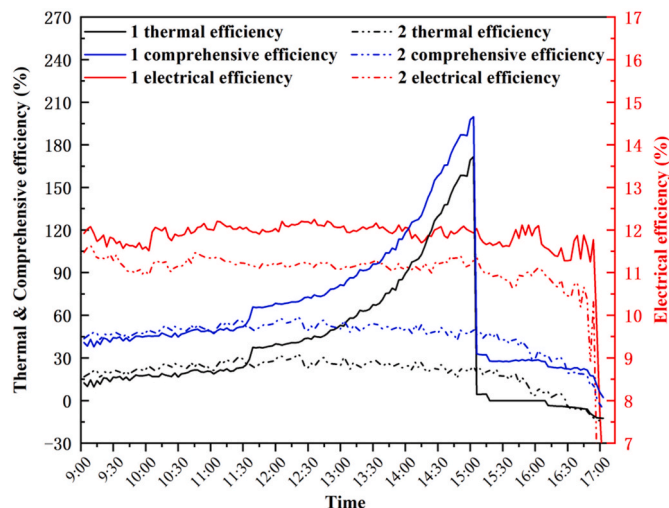


Fig. 13. The change chart of efficiency.

performance is more pronounced during the operation of the systems, i. e. it has a greater weighting. The average combined daily efficiency for System 1 is 66.36 % and the average combined daily efficiency for System 2 is 45.53 %, a growth of 45.75 % for System 1 over System 2. In this paper, there may be some differences between the experimental bench and the manufacturer's custom-made in terms of the connection and insulation of the heat pipe to the PV panel and the joint box. If the key components of the experimental system are custom-made by the manufacturer, the heat exchange and insulation will be tighter and the performance of the system can also be improved to some extent.

3.2.5. Pressure loss comparison

Fig. 14 shows the variation of pressure loss of the two systems. The fluctuation of the pressure loss of the two systems is very small, but the viscosity of the emulsion is larger than that of water, which in turn leads to an increase in the pressure loss of the system running emulsion. The pressure loss is approximately 5 kPa for the system running emulsion and 2.5 kPa for the system running water, and the energy required to operate system 1 is approximately twice that of system 2. The energy consumption caused by pressure loss in system 1 is calculated to be 7200 J. The heat gain of the system running emulsion is 1.35 MJ more than that of the system running water, which is 375 times of the total energy consumption of the system, and the energy loss of the system due to pressure loss is very small compared with the heat gain. Therefore, the application of paraffin/GO emulsion has a great role in improving the performance of the system.

This experiment utilized GO composite paraffin emulsion to enhance the thermal efficiency of the system by 92.28 % and electrical efficiency by 8.87 %. In order to clarify the effect of nanocomposite PCME used in this experiment on the significant performance enhancement of solar systems, some studies on the thermoelectric efficiency of PV/T system using other heat transfer media are summarized as shown in the Table 4. By comparing the experimental results, it can be found that other heat transfer media generally enhance the thermal efficiency of the system by no more than 70 %. Nanocomposite PCME is superior in the heat transfer and storage process of solar PV/T system compared to other heat transfer media.

4. Conclusions

The present study involved the preparation and analysis of paraffin/GO composite phase change material emulsions with variable GO

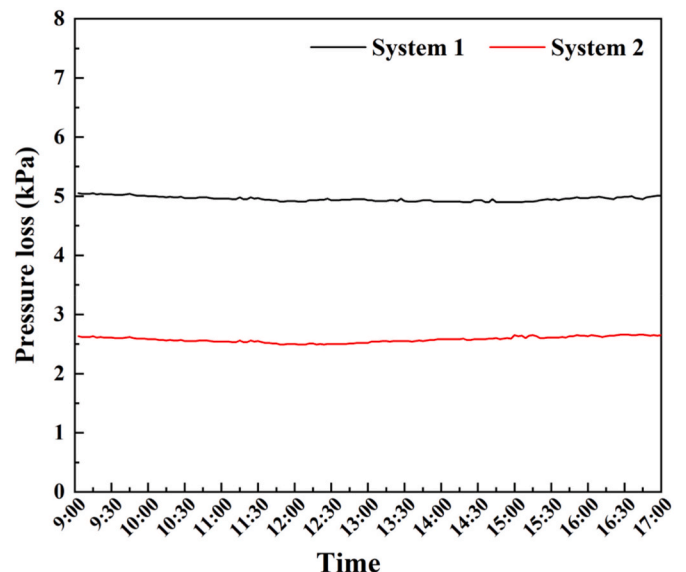


Fig. 14. The change chart of pressure loss.

Table 4

Investigations on the use of different fluids as heat transfer medium to improve the thermoelectric efficiency of solar energy systems.

Years	Authors	Medium		Efficiency enhancement		Systems
		Base fluid	Nano	Thermal/%	Electrical/%	
2022	Venkatesh [35]	water	Graphene nanoplatelet	23	13	PV/T
2021	Sheikholeslami [12]	water	MWCNT, Al ₂ O ₃ , CuO	29, 9, 20-30	–	flat plate solar collector
2021	Sangeetha [13]	water	MWCNT, Al ₂ O ₃ , TiO ₂	–	47, 33, 27	PV/T
2020	Tong [36]	water	MWCNT	8.8	–	flat plate solar collector
2016	Said [37]	water	Al ₂ O ₃	27.51-52.79	–	flat plate solar collector
2021	Fu [27]	water	MPCM	13	0.8	PV/T
2020	Samir [28]	Micro-encapsulated Phase Change Materials emulsion (MPCME)		68.8	14.1	BIPV/T
2015	Qiu [38]	MPCME		59	8.7	PV/T
2021	Feng [29]	Paraffin emulsion		–	5.8	PV modules
2015	Fiorentini [39]	Air with PCM		45	9	PV/T

contents of 0.01, 0.02, and 0.03 wt% for stability, latent heat, thermal conductivity, and viscosity. In addition, a comparison was made between the paraffin/GO emulsion with 0.02 wt% GO and water as heat transfer media for a flat-plate heat pipe solar PV/T system, and their effect on the thermoelectric performance of the system was evaluated. Several conclusions can be drawn from the study.

- (1) The addition of nanoparticles can improve the thermal conductivity of PCMEs, especially during the phase change. When the temperature is at 40 °C, the addition of 0.01, 0.02 and 0.03 wt% GO increases the thermal conductivity of paraffin emulsion by 5 %, 8 % and 10 %.
- (2) By comparing the four emulsions prepared in respect to the stability, thermophysical properties and viscosity, it can be found that the paraffin/GO emulsion with the addition of 0.02 wt% GO has better overall performance.
- (3) In the same test environment, the solar PV/T system running paraffin/GO emulsions had an average all-day thermal efficiency improvement of 92.28 %, an average electrical efficiency improvement of 8.87 % and an average overall efficiency improvement of 45.75 % compared to the system running water.

However, during the operation of the PV/T system, the mass flow rate of the emulsion in the system is difficult to be controlled due to the real-time variation of the density of the phase-change emulsion, which leads to a certain impact on the system performance, and the subsequent experiments can be further investigated from this aspect.

CRedit authorship contribution statement

Hongbing Chen: Writing – review & editing, Investigation, Conceptualization. **Xuening Gao:** Writing – original draft, Data curation. **Congcong Wang:** Writing – review & editing, Supervision. **Lizhi Jia:** Writing – review & editing. **Rui Zhao:** Investigation. **Junhui Sun:** Methodology. **Meibo Xing:** Writing – review & editing. **Pingjun Nie:** Writing – review & editing.

Declaration of competing interest

We declare that no conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication.

Data availability

Data will be made available on request.

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